Regis University
College for Professional Studies
School of Computer & Information Sciences

SYLLABUS

COURSE NUMBER: MT 360B

COURSE TITLE: Calculus II

Course Description Continues treatment of single variable calculus including definite and indefinite integrals, applications of integrals, transcendental functions, techniques of integration and infinite series.

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Course Prerequisites MT 360A, Calculus I

Course Outcomes

Upon successful completion of this course, students should be able to:

1. Demonstrate that the student is able to communicate mathematical concepts so that they may provide the public with information and expert judgment as part of the basis for social and political decisions.
2. Demonstrate an awareness of the socially beneficial uses of mathematics, as well as harmful uses of mathematics.
3. Calculate the area under a curve and volume of a solid by various methods.
4. Set up integrals to evaluate arc length and arc surface area.
5. Evaluate definite and indefinite integrals using a variety of tools.
6. Evaluate limits of indeterminate forms using L'hopital's rule.
7. Manipulate sequences and series according to mathematical rules.
8. Evaluate Maclaurin and Taylor polynomials and apply them.
9. Manipulate and graph polar equations.
10. Determine the solution of a first order separable differential equation.

Course Materials and Resources, including Technology Tools

Required:

- Hass, J., Weir, M. D., & Thomas, G. B. (2012). *University calculus: Early transcendentals (2nd ed.).* Upper Saddle River, NJ: Addison Wesley (Pearson). ISBN13: 978-0-321-75990-0 or ISBN10: 0-321-75990-7 hardcover (book w/MyMathLab).

- TI-8x or equivalent graphing calculator.

Method of Instruction

This course will use MyMathLab lab exercises and facilitated discussions.

Related Standards

Related Standards	
	This course enables students to achieve by the time of graduation:
	An ability to apply knowledge of computing and mathematics appropriate to the discipline
	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices
Regis Nine Educational Outcomes	
	Content Knowledge in Discipline
	Diverse Cultures, Perspectives, and Belief Systems
	Arts, Sciences, and Humanities
	Critical Thinking
	Communication
	Technology
	Ethical and Social Responsibility
	Leadership in Service to Others
	Lifelong Learning
Jesuit Themes - These are themes that are embedded as part of our interactions and instruction at Regis, but are not written in as activities	
	Magis, Finding God in All Things, Unity of Hearts and Mind, Contemplatives in Action, Care of the Whole Person, Men and Women for Others

Course Overview Grid

Week	Learning Topics	Activities	Reading Assignments	Assignments
1	Integration and Integrals • Indefinite Integrals • Substitution Rule for Indefinite Integrals • Substitution and Area Between Curves •	Discussion Forum: Topics 1 - 4 Lab Exercises (MyMathLab)	Chapter 5 Sections 5.5-6	First week assignment MyMathLab exercises (solve as assigned) Week 1 Quiz

Week	Learning Topics	Activities	Reading Assignments	Assignments
2	Applications of Definite Integrals • Volumes by Slicing and Axis Rotation • Volumes of Solids of Revolution / Method of Cylinders • Lengths of Plane Curves • Areas of Surfaces of Revolution • Work	Discussion Forum: Topics 1 - 3 Lab Exercises (MyMathLab)	Chapter 6, Sections 1-6	MyMathLab exercises (solve as assigned) Week 2 Quiz
3	More Applications of Definite Integrals • Logarithm Defined as in Integral • Separable Differential Equations • Exponential Growth • Hyperbolic Functions	Discussion Forum: Topics 1 - 4 Lab Exercises (MyMathLab)	Chapter 7 Sections 1-3	MyMathLab exercises (solve as assigned) Week 3 Quiz
4	Techniques of Integration – Part A • Integration by Parts • Trigonometric Integrals • Trigonometric Substitutions • Integration of by Partial Fractions	Discussion Forum: Topics 1 - 3 Lab Exercises (MyMathLab)	Chapter 8, Sections 1-4	MyMathLab exercises (solve as assigned) Week 4 Quiz
5	Techniques of Integration - Part B • Integral Tables • Numerical Integration • Improper Integrals • Comparison Test for Improper Integrals	Discussion Forum: Topics 1 - 4 Lab Exercises (MyMathLab)	Chapter 8, Sections 5-7	MyMathLab exercises (solve as assigned) Week 5 Quiz
6	Infinite Sequences and Series – Part A • Sequences • Infinite Series • The Integral Test	Discussion Forum: Topics 1 - 4	Chapter 9, Sections 1-6	MyMathLab exercises (solve as assigned) Week 6 Quiz

Week	Learning Topics	Activities	Reading Assignments	Assignments
	• Comparison Tests • The Ratio and Root Tests • Alternating Series, Absolute and Conditional Convergence	Lab Exercises (MyMathLab)		
7	Infinite Sequences and Series – Part B • Power Series • Taylor Series • Maclaurin Series • Convergence of Taylor Series • Binomial Series	Discussion Forum: Topics 1 - 4 Lab Exercises (MyMathLab)	Chapter 9, Sections 7-10	MyMathLab exercises (solve as assigned) Week 7 Quiz
8	Polar Coordinates and Conics • Polar Coordinates • Graphing in Polar Coordinates • Areas and Lengths in Polar Coordinates • Conic Sections • Polar Coordinates and Conics • Conics and Parametric Equations - The Cycloid	Discussion Forum: Topics 1 - 4 Lab Exercises (MyMathLab)	Chapter 10, Sections 1-6	MyMathLab exercises (solve as assigned) Final Exam

Student Evaluation Grid

Assignment	Value (percent of overall course grade)
Weekly Homework for Grade(8 in nuber)	30%
Weekly Quizzes (six in number)	30%
Midterm Exam	15%

Final Exam	15%
Forum Participation	10%
TOTAL	**100 %**

Participation

Because of the accelerated nature of the course, class participation is very important. Class participation/effort is important because we can all learn from each other. Your participation points can make a difference in the final grade. If the student doesn't participate during any given week, they will lose the participation points of that week.

Participation means:

1. Present in the forum every week.
2. Regularly checks forum and post all required assignments/discussion questions/ items by the deadlines.
3. Contributes to forum discussions.

Regis University Grading Scale

A	93 – 100	C	73 - 76
A-	90 – 92	C-	70 – 72
B+	87 – 89	D+	67 – 69
B	83 – 86	D	63 – 66
B-	80 – 82	D-	60 – 62
C+	77 – 79	F	59 or Lower

Attendance/Participation
Facilitator will provide specific attendance/participation policies in accordance with the official attendance policy of the School of Computer and Information Sciences.

Attendance Policy of the School of Computer and Information Sciences
Student participation is essential. Students are responsible for all material covered in each week. Students' interaction is central to the learning experience and imparts critical observations and knowledge to all concerned.

If a student misses more than one week of class participation without prior faculty approval, the facilitator may assign a failing grade.

Late Assignments/Assignment Revisions
Facilitator will provide specific late assignments/assignment revisions policies not to exceed the maximum official policy of the School of Computer and Information Sciences.

Late Assignments/Assignment Revisions Policy of the School of Computer and Information Sciences

Occasionally, work, health, and family obligations have priority over course work. If you experience these issues, please notify the facilitator in advance. If an assignment is due, please make every effort to submit your assignment on time. In the case of a negotiated absence or delay, assignments turned in by the agreed upon due date will receive full credit.

If you have not negotiated with the facilitator, assignments turned in late will be graded, then reduced by 25%. Assignments will not be accepted later than one week after due date.

Incomplete Grade
An incomplete grade is given at the discretion of the facilitator. A facilitator is not required to assign an incomplete grade. Each occurrence will be negotiated on an individual basis.

Incomplete Grade Policy of the College of Professional Studies
A grade of Incomplete or 'I" denotes that the required work for the course is incomplete due to unforeseen circumstances. Unforeseen circumstances mean, for example, that an accident, an illness, a death, or a major life transition has occurred. This grade is awarded at the discretion of the instructor and is submitted with an alternate grade of "I/F."

A student must submit a written request to the instructor asking that an incomplete grade be assigned prior to the end of the term in which the course is taken. The course instructor will determine the length of time for course completion, and the remaining requirements to complete the course; however the maximum time for completion is the end of the following semester.

Faculty will notify students in writing of the approval/denial of their request and provide instructions and guidelines required to resolve the incomplete grade. If the course work is not completed by the end of the next applicable semester and a grade change form submitted by the instructor, the incomplete grade reverts to the alternate grade assigned by the instructor and is calculated in the grade point average.

Disability Statement:
Equal access to education means equal opportunity to learn in an academic setting. Under Section 504 of the 1973 Rehabilitation Act and the Americans with Disabilities Act of 1990, students with documented disabilities who are otherwise qualified have a right to equal access to classes at Regis University. If you have a documented disability requiring academic adjustments for this class, please contact the Director of the Office of Disability Services at **303-458-4941** for discussion of disability issues. Since accommodations are not retroactive, it is suggested that you contact the Disability Services Office as soon as possible; adequate lead-time may be required to receive accommodations, including review of appropriate documentation.

Academic Dishonesty
Regis University is committed to intellectual integrity in its academic pursuits. Sanctions may, therefore, be imposed by the College for Professional Studies faculty or schools for cheating (defined as using inappropriate sources of information on a test) or plagiarism (defined as presenting as one's own, the ideas, words, or products of another). Such sanctions may include a failing grade on the assignment, failure of the course, or expulsion of the student from the course or the department.